

Elevator Pitch

As a student, I've often found that going through research papers is overwhelming. These PDFs are long, filled with equations and technical jargon, and it can take hours just to extract the main points. I've personally struggled with this—spending too much time reading when all I really needed were the key insights. And the truth is, most existing tools don't really address this problem for scientific papers, which leaves a big gap.

To solve this, I'm developing *SummarIQ*, an AI-powered PDF summarizer application. I'm building an intelligent system that takes a research paper in PDF format, extracts the core ideas and equations, and generates a concise summary in just a few sentences. I've also designed a simple iOS demo app where users can upload a PDF, let it process, and instantly view the summary in a clean, minimal interface.

On the technical side, I'm combining Python libraries like PyPDF2 and pdfplumber for text extraction with pretrained NLP models such as T5 and BART for abstract summarization. For extractive summaries, I'm using methods like LexRank. And on the frontend, I'm creating a smooth iOS experience with SwiftUI. The whole setup is lightweight, efficient, and well-suited for handling academic PDFs.

My goal with SummarIQ is to make reading research papers faster and more accessible. With this tool, students, researchers, and educators can quickly grasp the essence of a paper without wading through every single detail. Looking ahead, I plan to expand the system to handle equations more deeply, summarize multiple papers at once, and even deploy it on-device using Core ML. Ultimately, this project reflects both my passion for AI and iOS development, and my drive to build tools that save time and make learning easier.